Q.4. Part A: Determine the Thermal voltage V_T at $25^{\circ}C$

The Formula for thermal voltage is

$$V_T = \frac{kT}{q}$$

Step 1. Convert the temperature to Kelvin

$$T_K = T_C + 273$$

$$T = 25 + 273 = 298K$$

Step 2: Boltzmann's Constant.

$$k = 1.38 \times 10^{-23} J/K$$

Charge of an electron:

$$q = 1.602 \times 10^{-19} C$$

Step 3: $V_T = \dfrac{(1.38 \times 10^{-23})(298)}{1.602 \times 10^{-19}}$

$$V_T = 0.02567 V$$

$V_T \approx 25.7\,mV$

Part B.

Step 1: calculate the exponent

$$\frac{V_D}{nV_T} = \frac{0.5}{2 \times 0.02567}$$

$$= \frac{0.5}{2 \times 0.02567}$$

$$\approx 9.739$$

Step 2: Substitute into the diode equation.

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1\right)$$

Given, $I_S = 40 \times 10^{-19} A$

$$I_D = (40 \times 10^{-9})(e^{9.739} - 1)$$

$$e^{9.739} \approx 16966.5$$

$$I_D = (40 \times 10^{-9})(16966.5 - 1)$$

$$= 0.000678\,A.$$ Ans $I_D = 0.678\,mA$